



DLMS — Testing Guide

App: <https://dlms-rose.vercel.app>

What it is: the Device Lifecycle Management System — a structured, auditable replacement for the `PCBA_Traceability.xlsx` sheet. Device records (PCBA-A / PCBA-B / HMI / shipment info), role-based access, full change history, CSV/Excel import-export, and an AI invoice-extraction pipeline.

Please work through the walkthrough first (~20 min), then pick up whichever deep-dive sections you have time for. Log anything broken, confusing, or slow — see [Reporting issues](#) at the bottom.

Getting access

1. Go to <https://dlms-rose.vercel.app/signup> and register with your **@quantumtx.com** email (other domains are rejected — that itself is a test, see below).
2. Confirm your email (check spam; if no email arrives, ping Reet).
3. **Ping Reet to activate your account** — new accounts are created inactive and can't log in until an admin activates them.
4. Tell Reet which role you want for testing. Default is **Engineer**. We need at least one person testing as **Viewer** and one as **Admin** to cover the permission matrix.

Role	Can do
Viewer	Read-only: browse devices, history-free detail view, analytics
Engineer	Everything Viewer can, plus create/edit devices, change status, import/export, assign devices, log service events, confirm drafts
Admin	Everything Engineer can, plus soft-delete, user management, vocabularies, subscribers, full audit log

The database currently has only a handful of devices. That's intentional — the walkthrough has you create data as you go. Don't worry about making a mess; that's the point.

Walkthrough (~20 minutes, as Engineer)

1. Login & dashboard

- ☐ Log in — you should land on the **Dashboard**.
- ☐ Metric tiles render: Total Records, Total Units, By Status, By Customer.
- ☐ Click a status badge tile — it should deep-link to `/devices` pre-filtered to that status.
- ☐ Recent Activity feed shows latest changes with who/when.

2. Create a device

- ☐ **Data** → **New device**. Fill in the required fields (PCBA-A S/N, HW Rev, BOM Rev, FW Ver, Status, Phase) — use a serial like `TEST-<yourname>-0001`.
- ☐ Submit → you should land on the new device's detail page.
- ☐ Create a second device with the **same PCBA-A S/N** → should be blocked as a duplicate, with a link to the existing device.

3. Batch creation (serial ranges)

- ☐ **Data** → **New device** again. In PCBA-A S/N enter a range:
`TEST-<yourname>-0100 to 0104`.
- ☐ A blue banner should appear: "5 devices will be created..." and the button changes to **Create 5 Devices**.
- ☐ Submit → you land on `/devices` with a green "5 devices created successfully" banner, and 5 new rows exist (qty 1 each).
- ☐ Try a reversed range (`0104 to 0100`) → clear error, nothing created.
- ☐ Try an ambiguous serial (`SN1` and `SN2`) → error saying it can't be auto-expanded.

4. Device list

- ☐ Search for your serial in the search box (press `/` to focus it — keyboard shortcut).
- ☐ Sort by clicking column headers (toggles asc/desc).
- ☐ **Double-click** a cell (Customer, Remarks, serials...) to edit inline. Enter saves, Esc cancels.

- ☐ Open **Filters**, set a Build Date range, then open **Presets** → save the current filters under a name → clear everything → re-apply your preset.
- ☐ Select a few rows → the bulk action bar appears. Try **Change Status/Phase** on your test devices and **Export Selected**.
- ☐ **Export CSV** and **Export Excel** from the toolbar — files should open cleanly with bilingual headers and respect your active filters.

5. Device detail

Open one of your test devices:

- ☐ **Details tab** — field groups render with bilingual (EN | 中文) labels.
- ☐ **Assigned Engineers card** — assign yourself. Then go back to `/devices` and hit the **My Queue** toggle: only your assigned devices show.
- ☐ **Service Log tab** — log a service event ("Replaced fuse", today's date). It appears in the timeline grouped under "Today".
- ☐ **Change History tab** — every edit you've made so far should be there with field-level before → after diffs, actor, and version.
- ☐ **Components tab** — per-board config and change timeline.
- ☐ **Device Succession** — paste another test device's ID into "Link replacement" → the two devices now show "Replaced by →" / "← Replaces" links. Try linking a device to itself → should be rejected.

6. Import

- ☐ **Data → Import**. Save this as `test-import.csv` and upload it:

```
PCBA-A S/N,PCBA-A HW Rev,PCBA-A BOM Rev,PCBA-A FW Ver,Status,Phase,Customer,Qty,Ship Date
IMP-<yourname>-001,B,1.2,2.0.1,In Stock,Production,Acme Rehab,1,05/07/2026
IMP-<yourname>-002,B,1.2,2.0.1,Shipped,Production,Acme Rehab,2,01/07/2026
IMP-<yourname>-003,B,1.2,2.0.1,NotARealStatus,Production,Acme Rehab,1,31/02/2026
```

- ☐ Preview should show **2 valid / 1 rejected** — row 3 rejected with reasons (unknown status, invalid date).
- ☐ Import → result banner reports imported/skipped/invalid counts.
- ☐ Upload the same file again → the 2 rows should now be **skipped as duplicates**, not re-

imported.

7. Invoice extraction (AI)

- ☐ **Data → Drafts → Upload Document.** Upload any invoice-like PDF or a photo/screenshot of one (PDF/JPG/PNG, max 25 MB).
- ☐ Watch the status: Pending → Extracting... → Review.
- ☐ The confirm modal opens pre-filled. **Low-confidence fields are highlighted in yellow** with the source quote from the document — check the highlighting makes sense.
- ☐ Correct a field, then either **Confirm & Create Device** (creates the record, jumps to it) or **Save for review** (parks it in **Data → Drafts** for later — open it there and **Promote**).
- ☐ Upload the exact same file twice → the second attempt should be handled gracefully (dedupe by file hash), not create a duplicate.

8. Analytics & traceability

- ☐ **Insights → Analytics** — KPIs and charts render; switch the 7d/30d/90d range; try Export Excel / Export PDF.
 - ☐ As Engineer you should see a **My Queue** panel at the bottom (admin sees Engineer Activity instead).
 - ☐ **Insights → Traceability** — pick a group-by dimension (e.g. PCBA-A FW Ver); counts should reconcile with the device list; the **View** → links drill into `/devices` pre-filtered.
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Deep-dive checklists

Auth & onboarding

- ☐ Signup with a non-`@quantumtx.com` email → rejected with a clear message.
- ☐ Signup with mismatched passwords → "Passwords do not match".
- ☐ Login before admin activation → blocked.
- ☐ Wrong password → sensible error, no crash.
- ☐ Log out / session expiry → protected pages redirect to `/login` ; admin pages as non-admin land on the "Access Denied" page.

Role-based access (needs a Viewer + an Admin)

As **Viewer**:

- ☐ Nav shows only Dashboard / Devices / Insights (no Data or Admin dropdowns).
- ☐ Device list has no New Row button, no checkboxes/bulk bar, no inline editing.
- ☐ Device detail: no Edit/Delete buttons, no assignment controls, no service-log form, no Change History tab (Components tab shows a "restricted" note instead of history).
- ☐ **The important part — test the server, not just the UI:** paste gated URLs directly while logged in as Viewer: `/devices/new`, `/import`, `/drafts`, `/admin/users`, and the export URL `/devices/export?format=csv`. All should deny (redirect or 401), not work.

As **Admin**:

- ☐ Device detail has a **Delete** button → confirm dialog says soft-delete → device disappears from the list but its history is preserved (check `/admin/audit`).
- ☐ **Admin → Users:** change a tester's role via the dropdown; Deactivate then Reactivate someone (role select should be disabled while inactive).
- ☐ **Admin → Vocabularies:** add a new Status (code + EN + 中文 labels) → it immediately appears in the device form's status dropdown and is accepted by import. Toggle it inactive → it disappears from selects.
- ☐ **Admin → Audit Log:** every mutation from your testing session is there — actor, action, table, changed fields. Nothing missing, nothing editable.
- ☐ **Admin → Subscribers:** add an email for the weekly digest, toggle it active/inactive.

Concurrency (two browser tabs)

- ☐ Open the same device's edit form in two tabs. Save tab 1, then save tab 2 → tab 2 must get a "modified by another user — reload" conflict, not silently overwrite.
- ☐ Same with inline editing on the list, and with a bulk status change racing an individual edit → the bulk result should report the conflict count.

Status transitions — please probe this hard

There's a rules engine that only allows certain status moves (e.g. Retired and Lost are terminal — nothing should move out of them). Known soft spots to attack:

- ☐ **Bulk change:** select devices and try Change Status — the dropdown should only offer legal moves for *all* selected devices, and the server should reject illegal ones.
- ☐ **Edit-form bypass:** open a device's full edit page and change status there — the dropdown is *not* filtered on that page. Can you save an illegal move (e.g. Retired → In Use)? If yes, that's a finding.
- ☐ **Inline-edit bypass:** same question via double-click editing on the list.
- ☐ **"In Stock" / "Under Repair" devices specifically:** we suspect the transition rules don't recognize these two status codes (internal naming mismatch). Symptoms would be: bulk status change *a/ways* failing for them, and/or their status dropdown offering *every* status. Please test both statuses both ways and note exactly what happens.

Service & warranty reminders

- ☐ A device with a ship date (or last service) **more than 180 days ago** should show the red **Overdue for service** badge on its detail page, be counted in the orange banner on `/devices`, and be filterable via that banner's link. Create one via import with an old Ship Date (e.g. `01/11/2025`) to trigger it.
- ☐ Logging a service event on an overdue device should clear the overdue state (the event date becomes the new baseline).
- ☐ Dashboard also has separate warranty/service-due-soon banners (7-day window) — if you can set a warranty expiry within a week, check they fire.

Input validation & abuse

- ☐ Dates: the form and import should reject `31/02/2026` and `2025-13-40` but accept both `DD/MM/YYYY` and `YYYY-MM-DD`.
- ☐ Qty: negative numbers and text rejected.
- ☐ Search box with odd characters (`' ; DROP TABLE , %` , emoji) → a polite validation message, never a crash or a 500.
- ☐ Chinese text in Remarks (multi-line) → preserved verbatim through create, edit, export, and re-import.
- ☐ Export a CSV where a Remarks value starts with `=SUM(A1)` → the exported cell must be escaped so Excel doesn't execute it.
- ☐ Import an XLSX over 10 MB → friendly error. Upload a .txt to invoice extraction → rejected.
- ☐ A serial range of more than 5000 units on `/devices/new` → rejected.

Look & feel

- ☐ Try it on a laptop at half-width / a tablet — tables should stay usable.
 - ☐ Keyboard shortcuts: `/` focuses search, `?` opens the shortcut help, `Esc` closes dialogs.
 - ☐ Toasts appear for success/failure on saves, imports, bulk actions.
 - ☐ Nothing feels slow: device list paging, search, analytics. Note anything that takes >2–3s.
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Reporting issues

For each issue, drop a note in the team channel (or a shared doc) with:

1. **Where** — URL + what you clicked
2. **Who** — your role (viewer/engineer/admin)
3. **Expected vs actual**
4. Screenshot if visual

Severity guide:  data loss / wrong data / permission bypass ·  feature doesn't work ·  confusing or ugly.

Thanks! — Reet